Notes: Dixit (JPE, 1989)

Entry and Exit Decisions under Uncertainty

Contents

1 Big picture

- **Research question.** How should a firm decide when to enter and when to exit a market when output prices are uncertain and entry or exit is costly to reverse?
- **Core idea.** Entry and exit are not simple net-present-value rules. Because the firm can wait before entering and can preserve the option to reenter by not exiting, both the idle firm and the active firm contain option value.
- **Main object.** The solution is a pair of trigger prices: a high price P_H that induces entry and a low price P_L that induces exit.
- **Main result.** Uncertainty widens the zone of inaction. The entry trigger is above the Marshallian full-cost threshold, and the exit trigger is below the Marshallian shutdown threshold.
- **Hysteresis.** A firm that enters after a favorable price shock will not necessarily exit when the price returns to its old level. The firm's action depends on its state, not only on the current price.
- **Quantitative message.** Hysteresis is large even when sunk costs and uncertainty are small. In the central numerical case, the entry trigger is 33 percent above full cost and the exit trigger is 24 percent below variable cost.
- **Applications.** The model applies to foreign market entry, natural-resource extraction, fuel switching, hiring and firing, worker job search, and other decisions with uncertainty and costly reversibility.
- **Economic takeaway.** The value of waiting can dominate static cost comparisons. A firm may rationally delay profitable entry and rationally continue operating at a current loss.

2 Incremental contribution of the paper

- **Real-options treatment of entry and exit.** The paper extends the option-pricing analogy from a one-time investment option to a two-state problem in which active and inactive states are options on each other.
- **Linked options.** Entering is like exercising an option to acquire an active project, but the active project itself contains the option to abandon. Exiting returns the firm to the idle state, which contains the option to enter again.
- **Analytical clarity.** Related mine-opening and consumption/portfolio models existed, but Dixit strips the environment down to a simple setting that yields transparent analytical inequalities and comparative statics.
- **Hysteresis beyond static sunk cost.** Static sunk costs already create a gap between full cost and variable cost, but the paper shows that ongoing uncertainty creates an additional, often larger, gap.
- **Sharp trigger inequalities.** The paper proves that $P_H > w + \rho k$ and $P_L < w - \rho \ell$, so uncertainty strictly expands the inaction region relative to Marshallian thresholds.

- **Numerical discipline.** The paper solves the nonlinear trigger equations for plausible parameter values and shows that the option-value component is economically meaningful.
- **Broader interpretation.** The model explains persistent import presence, delayed investment, delayed firing, and other inertial behavior without irrational expectations, monopoly power, or strategic manipulation.

3 Economic setting and intuition

3.1 Three sources of hysteresis

Dixit organizes the intuition around three increasingly rich cases.

- **Static sunk costs.** If the price is expected to stay fixed forever, an idle firm enters when price exceeds full cost, $w + \rho k$. An active firm exits when price falls below the operating-cost threshold adjusted for exit cost, $w - \rho \ell$. This gives a Marshallian inaction band.
- **Mean reversion.** If the price is temporarily high but expected to fall back, the firm requires a price above full cost before entry. If the price is temporarily low but expected to recover, the active firm waits longer before exit.
- **Ongoing uncertainty.** Even without mean reversion, volatility creates option value. At the static break-even price, waiting has positive value because the firm can enter after favorable news but avoid entry after unfavorable news.

Economic message. Hysteresis does not require behavioral inertia. It follows from optimal forward-looking behavior when actions are costly to reverse.

3.2 Why state matters

A firm's optimal action depends on whether it is currently idle or active.

- **Idle firm.** The idle firm owns an option to enter. Exercising this option destroys the value of waiting, so entry requires a price high enough to compensate for losing that option.
- **Active firm.** The active firm owns operating assets plus an option to exit. Exiting also destroys the option to avoid future reentry costs, so the firm may stay active even at current operating losses.
- **Same price, different actions.** For prices between P_L and P_H, an idle firm remains idle and an active firm remains active.

Interpretation. The band (P_L, P_H) is a region of optimal inertia. The firm does not change state unless the price crosses one of the two trigger boundaries.

4 Model and notation

4.1 Technology and costs

The firm has access to a technology that produces one unit of output per unit of time.

- P: output price.
- w: variable operating cost per unit of time.
- k: lump-sum entry cost required to become active.
- ℓ: lump-sum exit cost required to suspend operations. The model also allows $\ell < 0$, which represents partial recovery of the entry cost.
- ρ: discount rate or cost of capital.
- The firm is risk neutral in the baseline model and maximizes expected net present value.

Important assumption. If the firm exits, the investment immediately rusts or decays. Reentry later requires paying k again. This is what makes entry and exit dynamically linked.

4.2 Price process

The market price follows geometric Brownian motion:

$$\frac{dP}{P} = \mu\, dt + \sigma\, dz,$$

where dz is the increment of a standard Wiener process:

$$E(dz) = 0, \qquad E(dz^2) = dt.$$

- μ is the expected growth rate of the price.
- σ is the volatility of the price process.
- The convergence condition is $\mu < \rho$, so the expected discounted value of future revenue is finite.

4.3 Value functions

There are two value functions because the firm has two possible states.

- $V_0(P)$: value of an idle firm when the current price is P.
- $V_1(P)$: value of an active firm when the current price is P.

Over the inaction range, the idle firm receives no flow profit, so its required return equals its expected capital gain:

$$\frac{1}{2}\sigma^2 P^2 V_0''(P) + \mu P V_0'(P) - \rho V_0(P) = 0. \tag{1}$$

The active firm receives flow operating profit $P - w$, which gives:

$$\frac{1}{2}\sigma^2 P^2 V_1''(P) + \mu P V_1'(P) - \rho V_1(P) = w - P. \tag{2}$$

Economic message. These are asset-pricing equations for real assets. The idle asset pays no dividend, while the active asset pays the operating-profit dividend $P - w$.

4.4 General solution

The characteristic equation for the homogeneous part is

$$\frac{1}{2}\sigma^2 \xi(\xi - 1) + \mu\xi - \rho = 0.$$

It has one positive root $\beta > 1$ and one negative root $-\alpha < 0$, where $\alpha > 0$. The economically relevant value functions are

$$V_0(P) = BP^{\beta},$$

and

$$V_1(P) = AP^{-\alpha} + \frac{P}{\rho - \mu} - \frac{w}{\rho}.$$

- BP^{β} is the value of the option to enter.
- $AP^{-\alpha}$ is the value of the option to shut down.
- $P/(\rho - \mu) - w/\rho$ is the value of operating forever without ever shutting down.

4.5 Trigger prices and boundary conditions

Let P_H denote the high trigger for entry and P_L denote the low trigger for exit.

At entry, the idle firm pays k and obtains the active-firm value:

$$V_0(P_H) = V_1(P_H) - k,$$

with smooth pasting:

$$V_0'(P_H) = V_1'(P_H).$$

At exit, the active firm pays ℓ and returns to the idle state:

$$V_1(P_L) = V_0(P_L) - \ell,$$

with smooth pasting:

$$V_1'(P_L) = V_0'(P_L).$$

Interpretation. Value matching says the firm is indifferent at the trigger. Smooth pasting says that the value function has no exploitable kink at the optimal switching boundary.

5 Analytical results

5.1 The key hysteresis result

Define the incremental value of being active rather than idle:

$$G(P) = V_1(P) - V_0(P).$$

The switching conditions become

$$G(P_L) = -\ell, \qquad G(P_H) = k, \qquad G'(P_L) = G'(P_H) = 0.$$

The function $G(P)$ is tangent to the horizontal line $-\ell$ at the exit trigger and tangent to the horizontal line k at the entry trigger. This tangency structure is the content of Figure 1.

Dixit shows that uncertainty implies

$$P_H > w + \rho k \equiv W_H,$$

and

$$P_L < w - \rho\ell \equiv W_L.$$

- W_H is the Marshallian full-cost entry threshold.
- W_L is the Marshallian exit threshold, adjusted for the exit cost.
- $P_H > W_H$ means firms delay entry relative to the static rule.
- $P_L < W_L$ means firms delay exit relative to the static rule.

Economic message. Uncertainty widens the inaction region beyond the range created by sunk costs alone. This is the paper's central analytical result.

5.2 Why zero exit cost does not eliminate exit hysteresis

Even when $\ell = 0$, the exit trigger remains below w. The reason is that remaining active preserves the option to avoid paying the entry cost k again if future prices recover.

- If the firm exits now, it saves current operating losses.
- But it gives up the chance to continue cheaply if price recovers.

- Therefore, the firm tolerates some operating losses before shutting down.

Interpretation. Exit is costly even without a literal exit fee, because exit can destroy future flexibility.

5.3 Limiting cases

- **No sunk costs.** If $k \to 0$ and $\ell \to 0$, then P_H and P_L both converge to w. Without sunk costs, there is no hysteresis.
- **Only one sunk cost.** If only one of k or ℓ goes to zero while the other remains positive, both trigger inequalities generally remain strict.
- **Very high exit cost.** If $\ell > w/\rho$, the project is never abandoned. Still, the entry trigger remains finite.
- **Impossible reentry.** If $k \to \infty$, the option to reenter becomes worthless, and the exit trigger is the price low enough to justify abandoning an irreversible opportunity.
- **No uncertainty.** As $\sigma \to 0$, the triggers collapse back to the Marshallian thresholds W_H and W_L.

5.4 Comparative statics

- **Higher variable cost w.** Both P_H and P_L increase.
- **Higher entry cost k.** Entry becomes harder and exit becomes more cautious: P_H rises and P_L falls.
- **Higher exit cost ℓ.** The inaction band also widens because exit becomes less attractive.
- **Higher volatility σ.** The value of waiting increases. Entry is delayed and exit is delayed, widening the band.
- **Higher trend μ.** A more favorable expected price trend lowers both triggers: firms are willing to enter sooner and are also more willing to keep operating through bad current prices.
- **Higher interest rate ρ.** Both triggers tend to rise because future option values and future profits are discounted more heavily.

Important point. The derivatives with respect to k and ℓ become very large near zero. Thus a tiny amount of irreversibility can create a large inaction band when there is uncertainty.

6 Numerical results

6.1 Central parameter values

Dixit uses the following baseline values:

$$w = 1, \qquad k = 4, \qquad \rho = 0.025, \qquad \sigma = 0.10, \qquad \mu = 0, \qquad \ell = 0.$$

Because $\rho k = 0.10$, the static full-cost entry threshold is

$$W_H = w + \rho k = 1.1,$$

and the static exit threshold is

$$W_L = w = 1.$$

The numerical solution gives

$$P_H = 1.4667, \qquad P_L = 0.7657.$$

- The entry trigger is about 33 percent above the full-cost threshold $W_H = 1.1$.
- The exit trigger is about 24 percent below the variable-cost threshold $W_L = 1$.
- The option-value component of hysteresis is much larger than the static Marshallian gap between W_H and W_L.

Economic message. Even modest uncertainty and modest sunk costs generate a wide inaction band.

6.2 Dumping and temporary below-cost sales

The numerical results imply that a firm can rationally remain active even when price is far below full cost and even below variable cost.

- This can look like dumping if an exporter continues selling at a price below average cost.
- But in this model, below-cost sales need not reflect strategic predation or price discrimination.
- They can be the rational outcome of option value: the firm accepts current losses to preserve the option of profitable future operation.

Policy implication. Antidumping logic based purely on price below cost can misinterpret optimal behavior under uncertainty.

6.3 Sensitivity to sunk costs

Figure 2 varies k while holding the other baseline parameters fixed.

- The upper curve reports P_H/W_H, the entry trigger relative to full cost.
- The lower curve reports P_L/W_L, the exit trigger relative to the exit threshold.
- The curves move away from one sharply near $k = 0$, showing that hysteresis appears rapidly even for small sunk costs.
- At $k = 0.4$, the interest on sunk cost is only 1 percent of variable cost, but the entry price is about 15 percent above full cost and the exit price is about 13 percent below variable cost.

6.4 Sensitivity to volatility

Figure 3 varies σ.

- As volatility increases, the upper entry-trigger ratio rises and the lower exit-trigger ratio falls.
- Even $\sigma = 0.025$ generates economically important hysteresis.
- At that volatility, the entry price is about 10 percent above full cost and the exit price is about 11 percent below variable cost.

Economic message. A little uncertainty matters a lot because it makes the option to wait valuable.

6.5 Sensitivity to price trend

Figure 4 varies μ.

- A higher expected price trend lowers both trigger ratios.
- A favorable trend makes entry attractive at a lower current price.
- It also makes the active firm more willing to ride out temporary losses.
- The quantitative effect is milder than the effects of sunk cost and volatility in the numerical examples.

7 Extensions

7.1 Mean reversion, floors, and ceilings

The baseline price process is a random walk in logs. Dixit explains that other price processes can be handled with the same logic.

- **Mean reversion.** If prices tend to move back toward a long-run level, high current prices are less favorable for entry and low current prices are less unfavorable for exit. The inaction band generally widens.
- **Reflecting barriers.** Government-imposed floors or ceilings can also alter trigger prices. Sufficiently tight barriers can eliminate entry or exit altogether.
- **Poisson jumps.** Jump processes provide another way to model discrete changes or probabilistic reversals.

7.2 Variable scale of output

If output can be varied after investment, flexibility changes but does not eliminate the logic.

- If there is no fixed cost or minimum scale, the firm can operate at an infinitesimal scale and may never need to abandon.

- With a flow fixed cost or a minimum efficient scale, exit remains meaningful.
- A model with a Cobb-Douglas operating technology can be transformed into a problem similar to the baseline model.

Interpretation. Operating flexibility can lower the triggers, but the option-value logic remains.

7.3 Variable project scale and investment bands

A richer model could let the firm choose the scale s of the project and pay adjustment costs when changing s.

- The single entry and exit triggers become adjustment boundaries $P_H(s)$ and $P_L(s)$.
- Instead of a single inaction interval, the model produces a band of inaction for capital adjustment.
- Industry equilibrium versions would endogenize price and relate entry and exit to demand or exchange-rate uncertainty.

Economic message. The paper is a building block for later real-options models of capacity choice, capital reallocation, and labor demand.

7.4 Risk aversion

The baseline model is risk neutral. Dixit shows how to modify the equations using risk-adjusted required returns.

- With a CAPM-style adjustment, risk aversion effectively changes the drift term in the valuation equations.
- If the project's price risk is positively correlated with market risk, risk aversion acts like a lower effective μ.
- From the numerical trend results, a lower effective μ raises both P_H and P_L.

Interpretation. Risk aversion can reinforce entry delay. Its effect on exit depends on how it changes the relevant risk-adjusted valuation of future profits and options.

8 Identification and interpretation

8.1 What kind of paper this is

This is a theoretical paper, not an empirical identification paper. Its main contribution is a formal model and numerical demonstration of a mechanism.

- The assumptions are chosen for transparency: risk neutrality, geometric Brownian motion, a single project, and simple lump-sum switching costs.
- The reduced model is not meant to capture every institutional detail of trade, labor markets, or natural-resource extraction.

- The goal is to show how costly reversibility and uncertainty generate optimal inertia in many economic environments.

8.2 Interpreting the trigger prices

- P_H is not a break-even price. It is the price high enough to compensate for exercising the option to wait.
- P_L is not the price at which current operating profits become negative. It is the price low enough to compensate for giving up the option to continue without paying reentry costs later.
- The distance $P_H - P_L$ measures the width of the inaction band, not simply the magnitude of physical sunk cost.

8.3 What the model explains

- Persistent imports after exchange rates reverse.
- Delayed investment despite positive current NPV under static rules.
- Delayed exit despite current losses.
- Large adjustment costs inferred in investment and labor-demand behavior.
- The value of maintaining flexibility in product-market, labor-market, and resource-extraction decisions.

9 Figures

The paper contains figures but no empirical tables. The package extracts all four figures from the paper.

9.1 Figure 1: Determination of entry and exit triggers

Read:

- Figure 1 plots $G(P) = V_1(P) - V_0(P)$, the value advantage of being active rather than idle.
- The exit trigger P_L is where $G(P)$ is tangent to $-\ell$. At this point, the active firm is just indifferent between continuing and exiting.
- The entry trigger P_H is where $G(P)$ is tangent to k. At this point, the idle firm is just indifferent between waiting and entering.
- Smooth tangency is the economic expression of optimal switching: there is no profitable marginal delay or acceleration at the boundary.

Economic message: Entry and exit thresholds are determined jointly. The active and idle states are linked options, not independent static decisions.

9.2 Figures 2–4: Numerical comparative statics

Read Figure 2:

- Figure 2 varies the sunk entry cost k.
- The entry-trigger ratio P_H/W_H rises above one and the exit-trigger ratio P_L/W_L falls below one.
- The steep movement near $k = 0$ shows that small sunk costs can generate a large inaction band.

Read Figure 3:

- Figure 3 varies volatility σ.
- More uncertainty raises the value of waiting, so entry is delayed and exit is delayed.
- The inaction region widens quickly even at low volatility levels.

Read Figure 4:

- Figure 4 varies the price drift μ.
- A higher expected price trend lowers both trigger ratios.
- The effect is economically intuitive but smaller than the effects of sunk cost and volatility in the examples.

Economic message: The numerical figures show that the real-options wedge is not a theoretical curiosity. It is large for plausible parameter values.

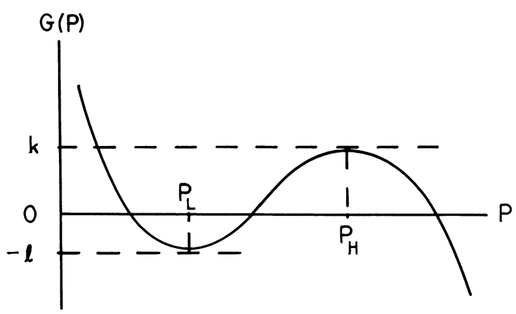

FIG. 1.—Determination of P_H and P_L

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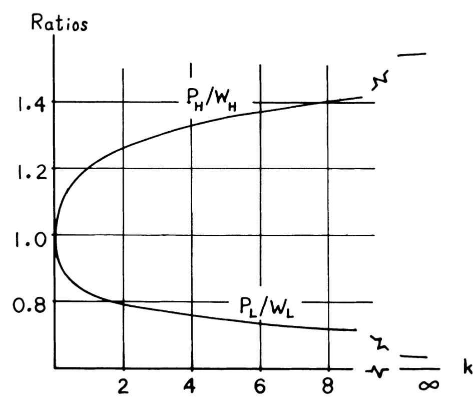

FIG. 2.—Effects of changes in k

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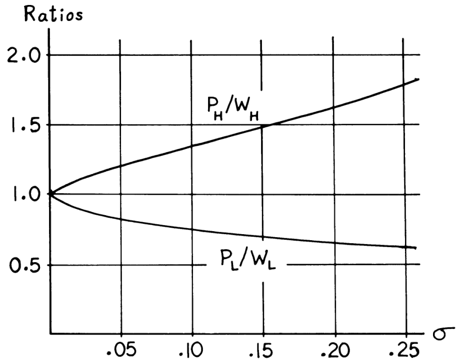

Fig. 3.—Effects of changes in σ

Figure 3: Effects of changes in σ.

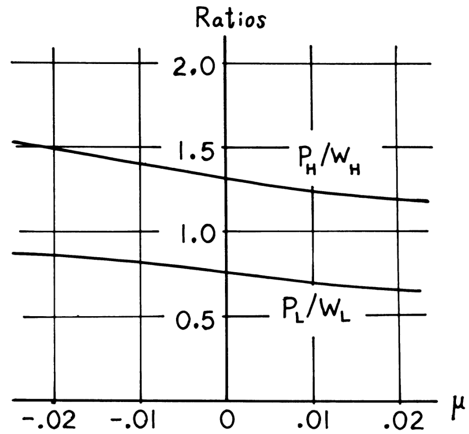

Fig. 4.—Effects of changes in μ

Figure 4: Effects of changes in μ.

10 Short summary

- The paper studies optimal entry and exit when output prices are uncertain and actions are costly to reverse.
- The firm has two states: idle and active. Each state is an option on the other.
- The price follows geometric Brownian motion in the baseline model.
- The optimal policy is a pair of trigger prices: enter when $P \geq P_H$ and exit when $P \leq P_L$.
- The region between P_L and P_H is a band of inaction. Inside that band, an idle firm stays idle and an active firm stays active.
- Uncertainty strictly widens the Marshallian inaction band: $P_H > w + \rho k$ and $P_L < w - \rho \ell$.
- The central numerical example gives $P_H = 1.4667$ and $P_L = 0.7657$, compared with static thresholds 1.1 and 1.0.
- Hysteresis emerges rapidly even when sunk costs or volatility are small.
- The model explains persistence in trade, employment, resource extraction, and other settings without requiring irrationality or market power.
- Extensions with mean reversion, price barriers, variable scale, and risk aversion preserve the main real-options intuition.

11 Conclusion

11.1 Core conclusion

Dixit shows that uncertainty and costly reversibility create optimal inertia. Firms should not enter as soon as static net present value becomes positive, and they should not exit as soon as current operating profit becomes negative. Instead, they should wait until price crosses a high entry trigger or a low exit trigger.

11.2 Economic intuition

The key economic force is the value of waiting. An idle firm can wait for better information before committing sunk capital. An active firm can keep operating through temporary losses to avoid paying the entry cost again later. These option values make firms less responsive to small or temporary price movements.

11.3 Contribution revisited

The paper makes the real-options logic precise in a compact entry-exit model. It gives analytical inequalities, boundary conditions, and numerical examples that show why hysteresis can be quantitatively large.

11.4 Implications

- **Investment.** Positive static NPV is not sufficient for entry when waiting is valuable.
- **Exit.** Negative current operating profit is not sufficient for shutdown when reentry is costly.
- **Trade.** Persistent foreign-market participation after exchange-rate reversals can be rational.
- **Labor markets.** Hiring and firing thresholds can differ sharply because training, search, and adjustment costs are partly sunk.
- **Policy.** Observing firms sell below full cost does not by itself prove predatory or strategic behavior.

11.5 Limitations

The model is deliberately stylized. It assumes an exogenous price process, a single project, immediate rusting after exit, simple lump-sum costs, and risk neutrality in the baseline. Industry equilibrium, endogenous prices, strategic interaction, gradual depreciation, and richer adjustment costs require additional modeling.

11.6 Takeaway

The enduring takeaway is that irreversible decisions under uncertainty should be made with option values in mind. Hysteresis is not an anomaly; it is the natural outcome of rational choice when waiting and flexibility are valuable.